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## Detailed Course

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## Motivation and Context

### What is Data Science?

**Data Science** studies:

- Modeling real-world phenomena as numerical (quantitative) data
- Geometric and statistical properties of this data
- Properties of the spaces in which this data resides
- Data analysis and the development of tools for this analysis
- Underlying assumptions in the design of these tools

**Relation to Machine Learning:** Data Science studies the representation spaces in which Machine Learning operates.

#### Quick Review: Motivation

**Data Science = ML:** Study of representation spaces where ML operates.

**Pipeline:** Process  $\rightarrow$  Sensor  $\rightarrow$  Raw Data  $\rightarrow$  Representation (matrix in  $\mathbb{R}^D$ ).

Focus on **geometric AND statistical** properties.

### Typical Data Pipeline

**General Flow:** Process  $\rightarrow$  Sensor  $\rightarrow$  Raw Data  $\rightarrow$  Representation

**Concrete Examples:**

- Camera  $\rightarrow$  image pixels  $\rightarrow$  matrix
  - Population  $\rightarrow$  survey results  $\rightarrow$  matrix
  - Text documents  $\rightarrow$  word occurrences  $\rightarrow$  matrix
  - Social network  $\rightarrow$  relationships  $\rightarrow$  matrix
  - Environment  $\rightarrow$  measurements  $\rightarrow$  matrix
- 

## Representation Spaces

### Formal Definition

We obtain data  $X = \{x_1, \dots, x_N\}$  where  $x_i \in \Omega \subseteq \mathbb{R}^D$  for all  $i \in \llbracket N \rrbracket$ .

**Statistical Perspective:**

- $X$  is a sample of size  $N$  from the underlying probability density function  $f_X : \Omega \rightarrow \mathbb{R}^+$  with  $D$  variables
- We declare  $N$  i.i.d. random variables  $\{X_i\}_{i \in [N]}$  where  $X_i \sim f_X$
- Each data point  $x_i$  is a realization (sample) of  $X_i$

#### Algebraic Perspective:

- We form the matrix  $X = \begin{bmatrix} | & & | \\ x_1 & \cdots & x_N \\ | & & | \end{bmatrix} \in \mathbb{R}^{D \times N}$
- The columns  $X_{\cdot j} = x_j$  are the data vectors

#### Space Properties

The representation space  $\Omega$  is generally a subset of  $\mathbb{R}^D$ :

- It is a **vector space** that can be equipped with an inner product  $\langle \cdot, \cdot \rangle$
- The inner product induces a **norm**  $\| \cdot \|$
- The norm induces a **distance function**  $d(\cdot, \cdot)$
- Favorable case:  $(\Omega, d)$  is a **metric space** where learning can be performed

**Note:**  $\Omega$  is also the basis of a measurable space on which random variables can be defined.

#### Quick Review: Representation Spaces

$X = \{x_1, \dots, x_N\}$  with  $x_i \in \Omega \subseteq \mathbb{R}^D$ .

**Two perspectives:**

- **Statistical:** Sample from  $f_X$ , i.i.d. variables
- **Algebraic:** Matrix  $X \in \mathbb{R}^{D \times N}$

**Structure:** Vector space  $\rightarrow$  inner product  $\rightarrow$  norm  $\rightarrow$  distance  $\rightarrow$  metric space.

#### Fundamental Questions

**What happens when:**

- $D$  increases? (dimensionality)
- $N$  increases? (number of samples)

**Impact on:**

- Data modeling
- Machine learning

### Curse of Dimensionality

#### 1. Data Sparsity

##### Density Estimation Problem

**Context:**

- Population on hypercube  $\Omega = [a, b]^D$
- Sample  $X$  from this population
- Objective: Estimate the empirical pdf via histogram

**Method:**

- Uniform quantization:  $b$  bins per dimension
- Desired quality:  $n$  samples per bin on average

**Question:** How many samples  $N$  are needed?

**Answer:**

- $D = 1 \rightarrow N \approx n \cdot b$
- $D = 2 \rightarrow N \approx n \cdot b^2$
- General  $D \rightarrow N \approx n \cdot b^D$

**Catastrophic Example:**  $b = 10, n = 10, D = 6 \rightarrow N \approx 10^7$  samples!

##### Estimation Quality with Fixed $N$

If we fix the number of samples at  $N$ :

**Question:** What estimation quality can we expect?

- $D = 1 \rightarrow n = \frac{N}{b} \rightarrow \mathbb{E}[P(x_i \in \text{bin}_j)] \approx \frac{1}{b}$
- General  $D \rightarrow n = \frac{N}{b^D} \rightarrow \mathbb{E}[P(x_i \in \text{bin}_j)] \approx \frac{1}{b^D}$

**Consequence:** Most bins are likely empty

##### Quick Review: Data Sparsity

To estimate density with quality, need  $N \approx n \cdot b^D$  samples (**exponential** growth).  
If  $N$  is fixed:  $n = N/b^D \rightarrow$  most bins **empty**.

Example:  $D = 6$ ,  $b = 10$ ,  $n = 10 \rightarrow N \approx 10^7$ .

## 2. Empty Sphere and Spiky Cube

### Volume of the Hypersphere

**Formula** for a hypersphere  $B_D(r)$  of radius  $r$  centered at the origin:

$$\text{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D\Gamma(D/2)}$$

**Recurrence Relation** (for  $D > 1$ ):

$$\text{vol}(B_D(r)) = \frac{2\pi r^2}{D} \text{vol}(B_{D-2}(r))$$

**Important Property:**

$$\frac{\text{area}(B_D(r))}{\text{vol}(B_D(r))} = \frac{\pi^{1/2} D\Gamma(D/2)}{\Gamma(\frac{D+1}{2})}$$

**Question:** How does the volume of the sphere evolve as  $D$  increases?

### Empty Sphere Phenomenon

**Configuration:** Hypersphere  $B_D(r)$  inscribed in hypercube  $C_D(r) = [-r, r]^D$ .

**Draw  $N$  uniform samples** from the cube (i.e., from  $\mathcal{U}(C_D(r))$ ).

**Question:** What proportion of samples falls within  $B_D(r)$ ?

**Calculation:**

- $\text{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D\Gamma(D/2)}$
- $\text{vol}(C_D(r)) = (2r)^D$

**Ratio:**

$$\frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = \frac{\pi^{D/2}}{D \cdot 2^{D-1} \Gamma(D/2)}$$

**Asymptotic Limit:**

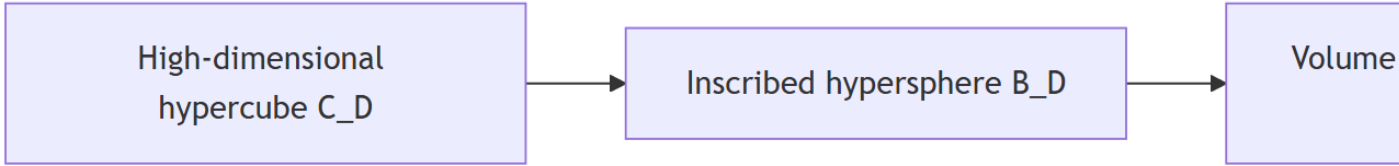
$$\lim_{D \rightarrow \infty} \frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = 0$$

**Note:**  $\Gamma(n)$  behaves like  $n! \sim n^n$

**Consequences:**

- All samples go to the part of the cube **outside the sphere** (the corners)
- The volume of the cube is **concentrated in its corners**
- All samples **escape from the center**
- The relative volume of the hypersphere tends to zero

**The sphere is likely empty**



**Corners of the Hypercube**

**Configuration:** Unit hypercube centered at  $C_D(1) = [-\frac{1}{2}, \frac{1}{2}]^D$  with  $\text{vol}(C_D(1)) = 1$ .

**Corner Properties:**

- Number of corners:  $n = 2^D$
- Each corner  $x_i$  has  $D$  coordinates:  $|x_i(j)| = \frac{1}{2}$  for all  $j$
- Distance to origin:  $\|x_i\|_2 = \frac{\sqrt{D}}{2}$

**Distance Table:**

| $D$                  | 1             | 2                    | 4 | 5                    | 100 |
|----------------------|---------------|----------------------|---|----------------------|-----|
| $\frac{\sqrt{D}}{2}$ | $\frac{1}{2}$ | $\frac{\sqrt{2}}{2}$ | 1 | $\frac{\sqrt{5}}{2}$ | 5   |

**Observation:** The distance to corners **increases with**  $\sqrt{D} \rightarrow$  corners move away from the center!

**Quick Review: Empty Sphere & Spiky Cube**

**Sphere/cube ratio:**  $\frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = \frac{\pi^{D/2}}{D \cdot 2^{D-1} \Gamma(D/2)} \rightarrow 0.$

Volume concentrated at **corners** of the cube (center-corner distance =  $\sqrt{D}/2$ ).

Uniform sampling  $\rightarrow$  **empty** sphere.

### 3. Volume Distribution

#### For the Hypercube

**Question:** How is volume distributed in a shape of size  $r$ ?

**Volume ratio** between hypercubes of side  $r$  and  $(1 - \varepsilon)r$  for  $\varepsilon \in ]0, 1[$ :

$$\frac{\text{vol}(C_D((1 - \varepsilon)r))}{\text{vol}(C_D(r))} = \frac{((1 - \varepsilon)r)^D}{r^D} = (1 - \varepsilon)^D$$

**Limit:**

$$\lim_{D \rightarrow \infty} \frac{\text{vol}(C_D((1 - \varepsilon)r))}{\text{vol}(C_D(r))} = 0$$

#### For the Hypersphere

**Volume ratio** between hyperspheres of radius  $r$  and  $(1 - \varepsilon)r$ :

$$\frac{\text{vol}(B_D((1 - \varepsilon)r))}{\text{vol}(B_D(r))} = \frac{2((1 - \varepsilon)r)^D \pi^{D/2}}{D \Gamma(D/2)} \cdot \frac{D \Gamma(D/2)}{2r^D \pi^{D/2}} = (1 - \varepsilon)^D$$

**Limit:**

$$\lim_{D \rightarrow \infty} \frac{\text{vol}(B_D((1 - \varepsilon)r))}{\text{vol}(B_D(r))} = 0$$

**Technical Note:**  $1 - \varepsilon \leq e^{-\varepsilon}$  so  $(1 - \varepsilon)^D \leq e^{-D\varepsilon} \rightarrow 0$  as  $D \rightarrow \infty$

### Generalization to Any Volume

**Principle:** Decompose the volume into arbitrarily small (hyper-)cubic voxels.

**Reduction:** If we reduce the length of a voxel side by factor  $(1 - \varepsilon)$ :

- Voxel volume reduced by  $(1 - \varepsilon)^D$
- Total volume reduced by  $(1 - \varepsilon)^D$

**Volume Ratio:**

$$\lim_{D \rightarrow \infty} \frac{\text{vol}(V_D(r)) - \text{vol}(V_D((1 - \varepsilon)r))}{\text{vol}(V_D(r))} = \lim_{D \rightarrow \infty} 1 - (1 - \varepsilon)^D = 1$$

**Fundamental Conclusion:** Volume is concentrated near the surface of the shape

#### Quick Review: Volume Distribution

For any shape: Volume ratio  $(1 - \varepsilon)^D \rightarrow 0$  as  $D \rightarrow \infty$ .

**Method:** Decomposition into hypercubic voxels.

**Result:** Volume is **concentrated at the surface** (difference  $\rightarrow 1$ ).

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## 4. The Gaussian Egg

### Density Along the Radius

**Question:** Given a Normal distribution  $\mathcal{N}(\mu, \Sigma)$ , what is its density along the radius  $r$ ?

**Context:** The coordinates are  $D$  i.i.d. random variables  $X_i \sim \mathcal{N}(0, 1)$ .

**Integration along  $r$**  = take the norm.  $R$  is the random variable associated with the radius:

$$R^2 = \|X\|_2^2 = \sum_{i=1}^D X_i^2 \sim \chi^2(D)$$

**Expectation:**

$$\mathbb{E}[R^2] = D \implies \mathbb{E}[R] \approx \sqrt{D}$$

We can consider the centered and scaled distribution (i.e.,  $\mathcal{N}(0, \text{Id}_D)$ ) with  $\sigma = 1$ .

**Density is concentrated at radius  $\sigma\sqrt{D}$**





### Gaussian Annulus Theorem

Given  $X \sim \mathcal{N}(0, \text{Id}_D)$ , for any  $\beta \leq \sqrt{D}$ :

$$P\left[\left|\|X\|_2 - \sqrt{D}\right| > \beta\right] \leq 3e^{-c\beta^2}$$

where  $c > 0$  is a constant.

**Interpretation:** The norm is **exponentially concentrated** around  $\sqrt{D}$ .

**Alternative Formulation:**

$$P\left[\left|\frac{\|X\|}{\sqrt{D}} - 1\right| \leq \varepsilon\right] \approx 1$$

Almost all density is in a **thin ring** around radius  $\sqrt{D}$

**Quick Review: Gaussian Egg**

For  $X \sim \mathcal{N}(0, \text{Id}_D)$ :  $R^2 = \|X\|_2^2 \sim \chi^2(D)$  so  $\mathbb{E}[R^2] = D$ .

**Gaussian Annulus Theorem:**  $P[|\|X\|_2 - \sqrt{D}| > \beta] \leq 3e^{-c\beta^2}$ .

Density concentrated at radius  $\sigma\sqrt{D}$  (**thin ring**).

## 5. Distance Concentration

**Context:** Data  $X \subset \Omega$ , select a data point  $x$  and its  $k$  nearest neighbors  $V_X^k(x)$ .

Let  $D_{\min}$  and  $D_{\max}$  be the random variables associated with the distances to the nearest and farthest neighbors among the  $k$  neighbors.

### Concentration Theorem

**Theorem:** If  $\lim_{D \rightarrow \infty} \text{Var}\left[\frac{d(x,y)^p}{\mathbb{E}[d(x,y)^p]}\right] = 0$  for  $0 < p < \infty$ , then for any  $\varepsilon > 0$ :

$$\lim_{D \rightarrow \infty} P\left[\frac{D_{\max} - D_{\min}}{D_{\min}} \leq \varepsilon\right] = 1$$

**Consequences:**

- All neighbors appear at distance  $D_{\min}$  from  $x$
- Discriminative power **decreases exponentially**
- KNN (*k-nearest neighbors*) and  $\varepsilon$ NN structures become **non-significant**

**Quick Review: Distance Concentration**

If normalized distance variance  $\rightarrow 0$ , then  $\frac{D_{\max} - D_{\min}}{D_{\min}} \rightarrow 0$ .

All neighbors are **equidistant**.

KNN and  $\varepsilon$ NN **lose their meaning**.

## 6. Quasi-Orthogonality

### Concentration of Angles Between Random Vectors

**Question:** When drawing vectors from a  $D$ -dimensional space, what is their expected angle?

**Method:** Declare 4 i.i.d. random variables  $W, X, Y, Z$  on  $\Omega \subset \mathbb{R}^D$  and compute:

$$\mathbb{E}[\cos(\angle(X - Y, Z - W))] = \mathbb{E}\left[\frac{\langle X - Y, Z - W \rangle}{\|X - Y\| \cdot \|Z - W\|}\right]$$

Since  $W, X, Y, Z$  are i.i.d.,  $X - Y$  and  $Z - W$  are also i.i.d. and centered:

$$\mathbb{E}(X - Y) = \mathbb{E}(Z - W) = 0$$

The numerator is  $\text{cov}(X - Y, Z - W) = 0$ , the denominator is a normalizer.

**Result:**  $\mathbb{E}[\cos(\theta)] = 0 \implies \theta \sim \frac{\pi}{2}$

**Alternative:**

$$\langle X - Y, Z - W \rangle = \langle X, Z \rangle - \langle X, W \rangle - \langle Y, Z \rangle + \langle Y, W \rangle$$

In expectation, all terms cancel because  $W, X, Y, Z$  are i.i.d.

**Conclusion:**

- Every random triangle is a **right triangle**
- Two random vectors are **quasi-orthogonal**

### Concentration of Angles in a Triangle

**Question:** What is the expected angle in a random triangle?

**Method:** Declare 3 i.i.d. random variables  $X, Y, Z$  and compute for an angle:

$$\mathbb{E}[\cos(\angle(X - Y, Z - Y))] = \mathbb{E}\left[\frac{\langle X - Y, Z - Y \rangle}{\|X - Y\| \cdot \|Z - Y\|}\right]$$

**Note:**  $X - Y$  and  $Z - Y$  are **no longer independent!**

**Calculation** (simplified):

$$\frac{\langle X - Y, Z - Y \rangle}{\langle X - Y, X - Y \rangle} = \frac{\langle X, Z \rangle - \langle X, Y \rangle - \langle Y, Z \rangle + \langle Y, Y \rangle}{\langle X, X \rangle + \langle Y, Y \rangle - 2\langle X, Y \rangle}$$

In expectation:

$$= \frac{\langle Y, Y \rangle - \langle X, Y \rangle}{2\langle Y, Y \rangle - 2\langle X, Y \rangle} = \frac{1}{2}$$

**Result:**  $\mathbb{E}[\cos(\theta)] = \frac{1}{2} \implies \theta \sim \frac{\pi}{3}$

**Conclusion:** Every random triangle is an **equilateral triangle**

### Impact for the Sphere

Given the unit ball  $B_D$ , if  $X$  is a random point on its surface and  $Y$  another point, let  $\theta = \angle(X, Y)$ :

- If  $Y$  sampled from  $\Omega$ :  $\mathbb{E}[\cos(\theta)] = \frac{1}{2} \implies \theta \sim \frac{\pi}{3}$
- If  $Y$  sampled from  $B_D$ :  $\mathbb{E}[\cos(\theta)] = 0 \implies \theta \sim \frac{\pi}{2}$

The volume of the sphere is contained at its **equator** (for any north direction  $X$ )

→ **Alternative proof** for volume concentration at the sphere's surface

**Note:** To sample on the unit hypersphere, sample from a centrally symmetric distribution (e.g., Gaussian) and normalize.

#### Quick Review: Quasi-Orthogonality

**2 random vectors:**  $\mathbb{E}[\cos(\theta)] = 0 \rightarrow \theta \sim \pi/2$  (quasi-orthogonal).

**Random triangle:**  $\mathbb{E}[\cos(\theta)] = 1/2 \rightarrow \theta \sim \pi/3$  (equilateral).

**Sphere:** Volume at the equator.

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## 7. Hubness

### Definition

**Question:** How many times does a sample appear as a  $k$ -nearest neighbor of another data point?

Given  $X$ , define:

$$r_{ik}(x_j) = \begin{cases} 1 & \text{if } x_j \in V_X^k(x_i) \\ 0 & \text{otherwise} \end{cases}$$

Then:

$$n_k(x_j) = \sum_i r_{ik}(x_j)$$

**Result:** The distribution of  $n_k$  values is **left-skewed**.

**Consequence:** A small number of samples appear in the neighborhood of **many** other samples.

These **hubs** are close to all other samples

**Empirical Visualization:** For 20-NN with  $D = 100$  and 1000 uniform samples on 50 bins, a strong asymmetry in the distribution of  $n_k$  is observed.

#### Quick Review: Hubness

$n_k(x_j)$  = number of times  $x_j$  is a k-NN of another point.

**Asymmetric** distribution: few **hubs** appear in many neighborhoods.

These hubs are “close” to all.

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## 8. Concentration Inequalities

### Fundamental Inequalities

**Markov’s Inequality:** For a non-negative random variable  $X$ :

$$P(X > \varepsilon) \leq \frac{\mathbb{E}[X]}{\varepsilon}$$

**Chebyshev’s Inequality:** For a random variable with mean  $\mu$  and finite variance  $\sigma^2$ :

$$P(|X - \mu| > \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$$

## Hoeffding's Inequality

**Statement:** If  $\{X_i\}_{i \in [D]}$  are independent random variables with  $P(a \leq X_i \leq b) = 1$  and  $\mathbb{E}[X_i] = \mu$ , then for any  $\varepsilon > 0$ :

$$P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$$

where  $\bar{X} = \frac{1}{D} \sum_{i=1}^D X_i$

**Note:** Example of PAC (*Probably Approximately Correct*) analysis: probability that an event occurs approximately.

## Explanation of Distance Concentration

Hoeffding's inequality explains:

- When aggregating independent variables, the aggregate **converges to their common expectation**
- The structure  $\sum_{i=1}^D (\cdot)$  corresponds to the calculation of Minkowski distances:

$$d_p(x, y) = \left( \sum_{i=1}^D |x[i] - y[i]|^p \right)^{1/p}$$

- Hoeffding's inequality formally explains why distance values **concentrate**
- Although counterintuitive that all neighbors are equidistant, it is intuitive that for large  $D$ , extreme values (small or large) of  $d_p(x, y)$  become unlikely: for  $d_p(x, y)$  to be small (large), **all**  $D$  coordinates  $x[i]$  and  $y[i]$  must be (dis)similar. As  $D$  increases, this becomes unlikely.

### Quick Review: Concentration Inequalities

**Hoeffding:**  $P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$ .

Aggregation of independent variables  $\rightarrow$  convergence to  $\mu$ .

Structure  $\sum_{i=1}^D (\cdot)$  in distances  $\rightarrow$  **concentration of distances**.

## Blessing of Dimensionality

### 1. Laws of Large Numbers

#### Weak Law of Large Numbers (WLLN)

**Statement:** Given a set of  $D$  i.i.d. random variables  $\{X_i\}_{i \in [D]}$  with  $\mathbb{E}[X_i] = \mu$ , then for any  $\varepsilon > 0$ :

$$\lim_{D \rightarrow \infty} P(|\bar{X} - \mu| < \varepsilon) = 1$$

where  $\bar{X} = \frac{1}{D} \sum_{i=1}^D X_i$

(Can be proven from Chebyshev's inequality)

**Interpretation:**

- The WLLN states that the **empirical mean**  $\bar{X}$  converges (in probability) to the **true expectation**  $\mu$  as the number of samples increases
- Fundamentally consistent with the frequentist approach to probabilities
- It is the **fundamental basis** of estimation theory (and density estimation)

#### Central Limit Theorem (CLT)

The form and rate of convergence are given by the CLT:

$$Z_D = \frac{\sqrt{D}}{\sigma}(\bar{X} - \mu) \sim \mathcal{N}(0, 1)$$

if  $\text{Var}(X_i) = \sigma^2$

#### Quick Review: Laws of Large Numbers

**WLLN:**  $\bar{X} \xrightarrow{P} \mu$  as  $D \rightarrow \infty$ .

Basis for density estimation.

**CLT:**  $Z_D = \frac{\sqrt{D}}{\sigma}(\bar{X} - \mu) \sim \mathcal{N}(0, 1)$  (convergence rate).

## 2. Random Projections

### Principle

**Recall:** Two random vectors  $u$  and  $v$  are likely orthogonal.

**Quasi-orthogonality:**  $\langle u, v \rangle = \varepsilon$  (small).

**Key Property:** **Exponentially many** quasi-orthogonal directions in high-dimensional spaces.

**Application:** Create random bases (quasi-orthonormal) for signal decomposition and coding.

### Johnson-Lindenstrauss Lemma

**Statement:** Given  $X \subset \Omega$ , there exists a linear map  $p : \mathbb{R}^D \rightarrow \mathbb{R}^d$  such that, with  $0 < \varepsilon < 1$  and for all  $i, j \in \llbracket N \rrbracket$ :

$$(1 - \varepsilon)\|x_i - x_j\| \leq \|p(x_i) - p(x_j)\| \leq (1 + \varepsilon)\|x_i - x_j\|$$

and

$$d = O\left(\frac{\log N}{\varepsilon^2}\right) \leftarrow \text{Does NOT depend on } D!$$

### Applications:

- *Locally Sensitive Hashing* (LSH) for approximate nearest neighbor (ANN) search
- **Caveat:** Convergence rate slower than most concentrations

#### Quick Review: Random Projections

**Principle:** Exponentially many quasi-orthogonal directions in high dimensions.

**Johnson-Lindenstrauss:** Projection  $\mathbb{R}^D \rightarrow \mathbb{R}^d$  preserving distances (up to  $\varepsilon$ ) with  $d = O(\log N / \varepsilon^2)$  (independent of  $D$ !).

Application: LSH for ANN.



## Importance in Data Science

### 1. Sampling

#### Consequences of Sparsity:

- An **exponential number of samples** is needed to cover a high-dimensional space
- Concentration inequalities (including WLLN and CLT) are **precision indicators** for density estimation
- In practical processes like rejection sampling, the **rejection rate** becomes exponentially close to 100% (cf. empty sphere)

### 2. Indexing

#### Exploiting Structure:

- Indexing exploits data structure to anticipate which parts of space  $\Omega$  to visit at query time
- If all neighbors appear at **equal distance** from the center, the structure **disappears**

#### Spatial Partitioning Techniques (e.g., *kd-trees*):

- Attempt to partition space so that each data point is uniquely identified within a cell intersection (paths in the search tree)
- With distance concentration, cell size reaches the **sphere's radius** to cover the entire space

Search becomes **exhaustive** complexity  $O(N)$

- *Hubness* makes some data points answers to **almost all queries**

### 3. Learning

#### Function Approximation

**Machine Learning** = discovering **function approximators**.

**Problem:** Given  $\phi : X \rightarrow Y$  the true function mapping data  $X$  to labels  $Y$ , ML seeks an approximation  $\phi_\theta \in \mathcal{F}$  of  $\phi$  minimizing error (risk) on samples (train/test).

## Theoretical Bound

If  $\mathcal{F}$  is a class of  $c$ -uniformly Lipschitz functions on  $\Omega$ :

$$\sup_{\phi_\theta \in \mathcal{F}} \|\phi_\theta - \phi\|_\infty \geq \frac{c\sqrt{D}N^{-1/D}}{2} \sqrt{\frac{2}{\pi e}} \left(1 + O\left(\frac{\log D}{D}\right)\right)$$

If we allow an error  $c\varepsilon$  for  $\varepsilon > 0$ , the required number of training examples  $N$  is:

$$N \geq \frac{\varepsilon^{-D} D^{D/2}}{(2\pi e)^{D/2}}$$

## Concrete Examples:

- $c = 1, \varepsilon = 0.1, D = 3 \rightarrow N \geq 74$
- $c = 1, \varepsilon = 0.1, D = 10 \rightarrow N \geq 687 \times 10^6$
- $c = 1, \varepsilon = 0.1, D = 15 \rightarrow N \geq 377 \times 10^{12}$
- $c = 1, \varepsilon = 0.01, D = 10 \rightarrow N \geq 6.9 \times 10^{18}$

## Realistic Case in ML:

- ML data: typically hundreds of features ( $D \sim O(10^2)$ )
- Typical desired error:  $\varepsilon \approx 10^{-4}$

$$N \geq \frac{10^{4 \cdot 100} \cdot 100^{50}}{17.07^{50}} \approx 10^{400}$$

**Conclusion:** This is **impossible** in practice!

### Quick Review: Importance in Data Science

**Sampling:** Exponential sample requirement, rejection rate  $\rightarrow 100\%$ .

**Indexing:** Structure disappears, kd-trees  $\rightarrow$  complexity  $O(N)$ , hubness.

**Learning:**  $N \geq \varepsilon^{-D} D^{D/2} / (2\pi e)^{D/2} \rightarrow$  impossible for large  $D$  and small  $\varepsilon$ .

#### 4. Why Does Data Science Still Work?

**Question:** Why do data science processes work despite these theoretical barriers?

**Assumptions in the Above Analysis:**

- Most analyses assume a **uniform** (or isotropic – Normal, ...) distribution of data
- Most analyses assume **independent** features

**Reality:**

- **Real data does NOT populate the feature space uniformly** (if features are informative)
- **Features show correlations**

In practice, data populates **subspaces** of  $\Omega$  with **lower local intrinsic dimensionality**

**Solutions:**

- **Decorrelate features** (e.g., via linear latent space)
- **Discover data subspaces** (clustering, nonlinear dimensionality reduction)

##### Quick Review: Why It Works

Analyses assume **uniform** distribution and **independent** features.

**Reality:** Data lies on **subspaces** of lower intrinsic dimensionality.

**Solutions:** Decorrelation, clustering, dimensionality reduction.

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#### Global Summary

**High-dimensional geometry** ( $D \gtrsim 10$ ) is **different** from familiar 3D geometry.

##### + Positive Aspects

- **Concentration inequalities** are the basis of statistical estimation
- **Laws of Large Numbers** enable density estimation
- **Random projections** exploit quasi-orthogonality

## - Negative Aspects (Curse)

- Data becomes very **sparse** unless using exponentially more data
- Even with dense data, **distance and angle values concentrate**
  - From a query, all neighbors are **equidistant**
  - From a direction, everything else appears **orthogonal**
- Density concentrates on **narrow radii** → biased sampling
- **Hubs** are part of all class/query responses

## Solution

Since this is by construction, the only solution is to operate in **low-dimensional spaces** in some way:

- Dimensionality reduction
- Subspace discovery
- Feature decorrelation

### Quick Review: Summary

**High dimension** ( $D \gtrsim 10$ ) 3D geometry.

: Concentration inequalities for estimation, random projections.

: Data sparsity, distance/angle concentration, hubs.

**Solution:** Operate in **low dimension**.